



SIMATS ENGINEERING

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Course Code: ITA0611

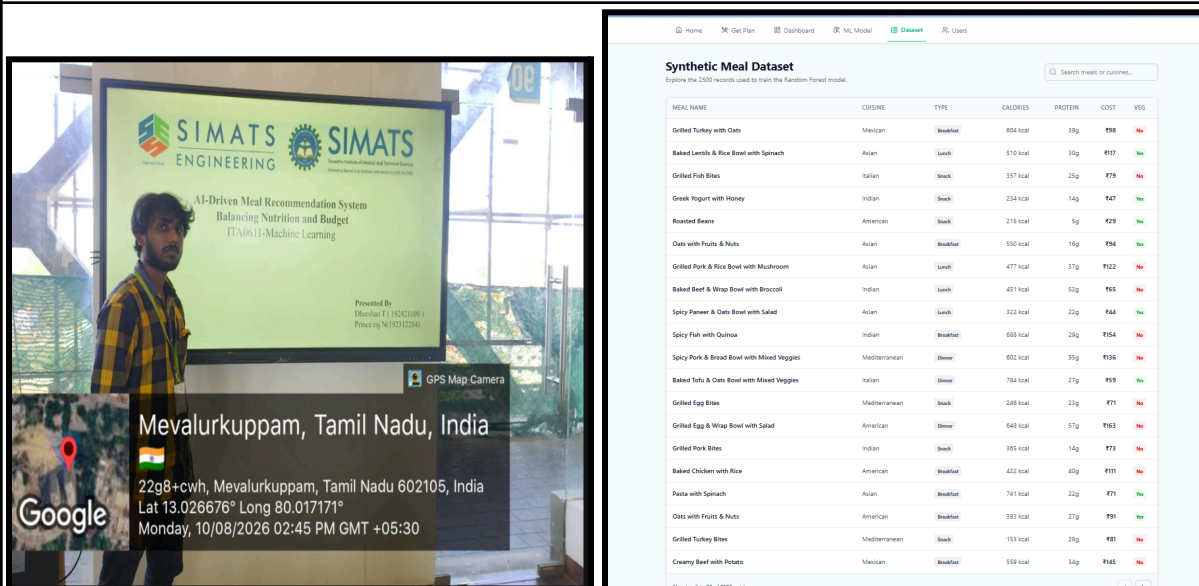
Slot: A

Course Name: Machine Learning

Course Faculty: Dr.P.Jasphin Jeni Sharmila

Project Title: AI-Driven Meal Recommendation System Balancing Nutrition and Budget

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: The AI-Driven Meal Recommendation System Balancing Nutrition and Budget is designed to recommend healthy and affordable meals based on user requirements. It considers factors such as calorie needs, nutritional goals, food preferences, and budget. The system analyzes food items based on their nutritional value and cost and uses AI-based recommendation techniques to select suitable meal combinations. It provides personalized breakfast, lunch, dinner, and snack recommendations within the user's budget. Overall, the system helps users maintain a balanced diet while reducing unnecessary food expenses.

My contribution focused on dataset handling, recommendation workflow, interface design, and system testing. I helped prepare and organize the food dataset with nutritional and cost-related information and contributed to developing the recommendation workflow based on nutrition and budget requirements. I also supported the design of a simple and user-friendly interface for entering requirements and viewing meal recommendations. I tested different inputs and budget conditions to verify that the system generated suitable results. I identified errors and usability issues and supported necessary improvements. This contribution helped improve the accuracy, usability, reliability, and overall effectiveness of the system.

Student Signature

Guide Signature